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Tut - 5.

1) $f(n) = 2^n$ $\quad g(n) = n!$ $\quad h(n) = n^{\log n}$

→

$$2^n < n!$$

$$n^{\log n} = e^{(\log n)^2}$$

$$n^{\log n} < 2^n < n!$$

$f(n) = O(g(n))$ $\quad g(n) = \Omega(f(n))$

$h(n) = O(f(n))$

2)

→ for (i = 1; i<n)

↳ Runs n times

for (j = n; j > 1; j = j/2)

Runs $p = \log n$ times

for (k = 1; k<p; k = k*2)

$\log(\log n)$

For each outer loop

$\log(\log n)$

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$n \log(\log n)$.

3) n Array $A_1, A_2, A_3 \ldots$
Median of Medians?

→ Each median $= O(n)$
for $n = n \times O(n) = O(n^2)$

Sort
$O(n \log n)$

$O(n^2) > O(n \log n)$ So. $O(n^2)$

4) for $(i = n, j = 0, i > 0, i = i/2, j{+}{+} = 1)$

→ Iterations
$i = i/2$
also
$\log n$.

NOTES

$j = n + \frac{n}{2} + \frac{n}{4} + \ldots$

$j = 2n$ hence $j = \Theta(n)$

$Val(j) = \Theta(n)$

5) $j = 1$
while $(j <= n)$
$j = j * 2$

$j = 1, 2, 4, 8 \ldots\ldots 2^k$

$2^k > n$

$k = \log_2 n$

Comparisions $= \Theta(\log_2 n)$

NOTES

6) Majority element

→ $A[i] = A[i + n/2]$

Constant checking required hence

$\Theta(1)$